

HomeView: Automatically Building Smart Home Digital Twins With Augmented Reality Headsets

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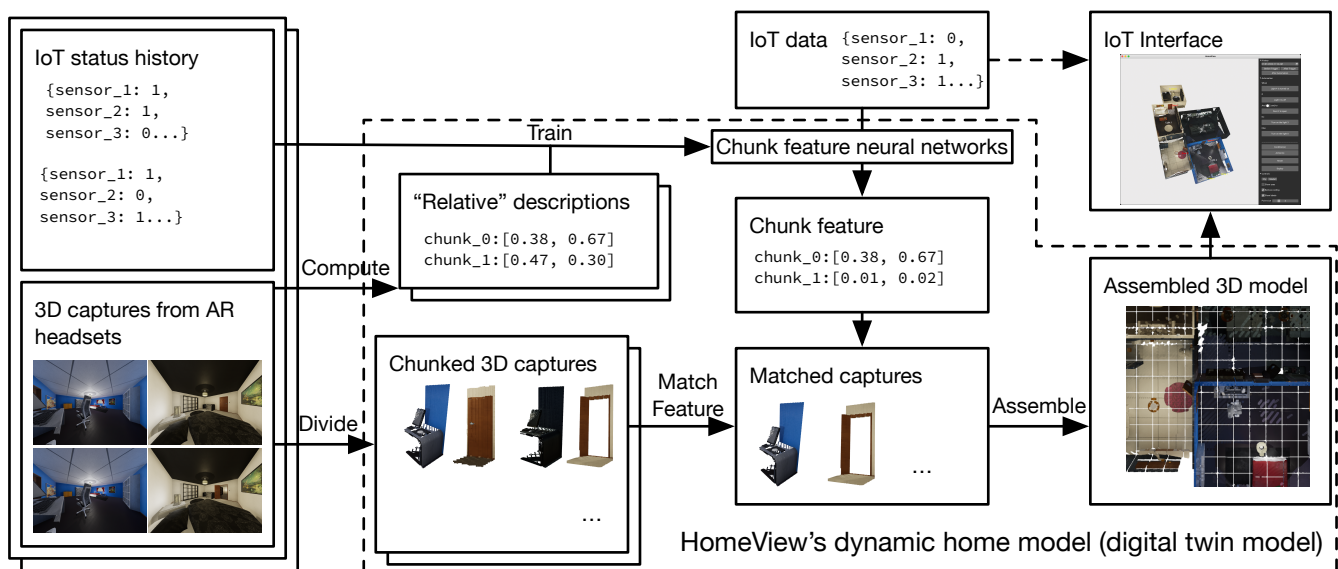


Figure 1: HomeView automatically generates a digital twin of the home using 3D captures from the AR headset and corresponding IoT data. It uses a divide-and-assemble approach (Section 3.1) to render unseen scenarios with a small amount of data.

ABSTRACT

Digital twins have demonstrated great capabilities in the industrial setting, but the cost of building them prohibits their usage in home environments. We present HomeView, a system that automatically builds and maintains a digital twin using data from Augmented Reality (AR) headsets and Internet of Things (IoT) devices. We evaluated the system in a simulator and found it performs better than the baseline algorithm. The user feedback on programming IoT devices also suggests that contextual information rendered by HomeView is preferable to text descriptions.

CCS CONCEPTS

- **Human-centered computing** → **Mixed / augmented reality**;
- **Computing methodologies** → **Interactive simulation**.

KEYWORDS

digital twins, internet of things, augmented reality

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1 INTRODUCTION

Digital twins are dynamic virtual representations of physical systems that enable better understanding, learning, and reasoning [7]. Therefore, building digital twins for smart homes could potentially

help users manage the Internet of Things (IoT) devices more easily. There are tools [1, 2] to build the digital twins, but they require experts to manually create models that match the environment. Besides, setting up IoTs is laborious [16], making the process even more prohibitive to ordinary users. Augmented Reality (AR) has been successfully used in IoT interactions [20, 23, 28] but often requires unique markers [15, 19], tags [43], or manual calibration [20].

We present HomeView¹, a system that automatically builds a digital twin for the smart home. HomeView takes colored point clouds from the AR as well as the corresponding IoT data as inputs, divides the point clouds into chunks, and learns to associate the IoT status with the chunks therefore automatically assembles a new point cloud (digital twin) for any given IoT status. The user can then use this digital twin to program the IoTs using the interface we provide (Section 3.4).

We evaluate the system in a simulator (Section 3.3), where the AR headset is represented by cameras set up in the simulator. Regarding visual fidelity, our approach correctly generates 94.68% of the chunks, while the baseline retrieve-from-history algorithm's accuracy is 12.25% (Section 4.1). We conducted a user study with five participants to perform IoT trigger-action programming [50]. Four preferred using HomeView with its context information than the baseline system with the textual one (Section 4.2).

The contributions of this research includes:

- (1) a system that automatically constructs digital twins of the environment using 3D captures from AR headsets and sensor data from IoTs;
- (2) an IoT programming interface using HomeView to present contextual information;
- (3) a preliminary evaluation of the HomeView system.

2 RELATED WORK

This section describes related works on digital twin modeling as well as IoT visualization and interaction in 3D.

2.1 Digital twin modeling methods

We categorize the existing methods into physics-based [18] and data-driven [35]. The former requires the experts to use simulation software, such as Ansys [1] and COMSOL [2], to describe the underlying physics rules for the model. The latter uses data from the real world and often uses machine learning to produce the model, which requires less manual labor. However, prior work [10, 29, 38, 49] focuses on metrics used in industry, such as reliability and efficiency, which is not that relevant to ordinary users' experiences. HomeView uses the data-driven approach but focuses on higher visual fidelity that provides users better understanding of the real world.

2.2 IoT visualization techniques

We categorize the existing techniques into spatial and temporal.

The spatial technique overlays the sensor data at its corresponding position on a 2D floor plan [6, 26, 41], which is incapable of visualizing the locations in the vertical dimension. Improvements such as including building information [12, 33], using 3D models [21] in VR [39, 40] and AR [25, 40] require manual setup and the

IoT data is often presented numerically. The temporal technique uses line graphs [11, 31] to represent the time-varying sensor output, which lacks the spatial information that is crucial for users to understand the real world.

HomeView leverages both techniques: it visualizes the given sensor data by rendering the corresponding digital twin without manual setup; It visualizes temporal data by playing a sequence of generated digital twins as a video.

2.3 IoT interaction in 3D

We categorize the existing approaches into *in situ* interaction and *remote* interaction.

in situ allows on-site viewing, control, and simulation of IoTs. Prior work overlays IoT data in virtual environments and enables users to interact through AR [24], touchscreen and virtual objects [14, 32, 36, 44, 48], as well as gestures and other modalities [22, 42, 47]. *remote* interaction takes place in a different space-time. For space, some [44] allow users to control devices from afar, while others [27, 30] enable bidirectional interactions between local and remote devices. For time, prior work allows users to preview the program for deployment [9] and view historical events to ensure the contextual triggers are set up correctly [46].

HomeView enables both *in situ* and *remote* interactions while not requiring manual visualization creation.

3 SYSTEM DESIGN

In this section, we describe HomeView's divide-and-assemble method, machine learning algorithm, simulator, and programming interface.

3.1 Divide-and-assemble method

Three methods of generating 3D models from IoT data are considered (Figure 2).

3.1.1 Synthesized. Directly synthesizing the 3D model from IoT data requires a massive amount of training data, because it generates high-dimensional space descriptions (point clouds) from low-dimensional IoT data. Moreover, the outcomes tend to have visual artifacts. (Figure 2 left).

3.1.2 Retrieve-from-history. Retrieving a scene from history requires less training data, but it only supports situations the algorithm has seen before. For n IoTs, there are 2^n situations. Since it is not practical for ordinary users to trigger all situations to train the system, the possible outputs are limited (Figure 2 middle).

3.1.3 Divide-and-assemble. This method is therefore proposed to accommodate the shortcomings of the previous two. We notice that each IoT has limited spacial influence. For example, a light only affects a subregion of the entire home. Conversely, a given subregion of the environment is dominated by a limited number of IoTs. Therefore, we can divide a space into chunks of equal size, where size is an adjustable parameter based on the given environment. We then train a neural network for each chunk to learn its relationships with IoT data, which can be used to assemble the chunks back into a 3D model given specific IoT status (Figure 2 right).

¹<http://jya.ng/homeview>

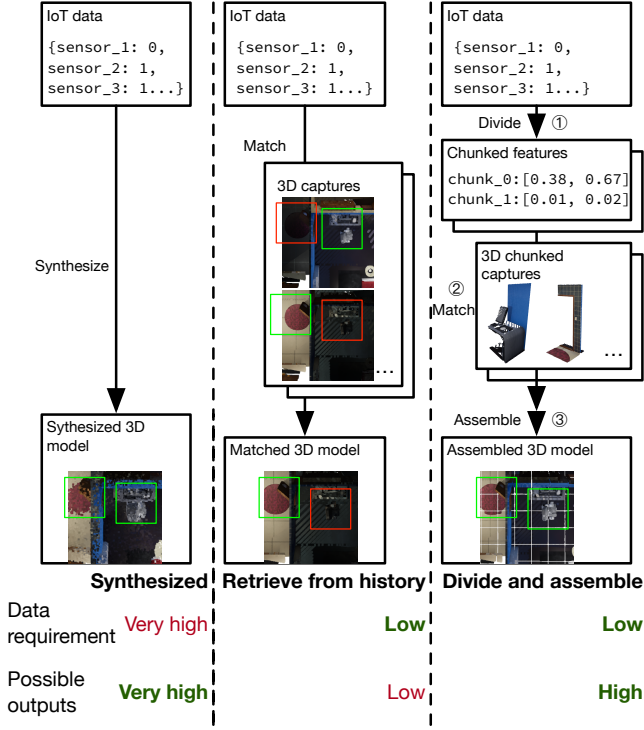


Figure 2: HomeView’s divide-and-assemble approach produces correct rendering of the carpet and the computer (green boxes), while the retrieve-from-history method returns an incorrect rendering of the computer (red boxes) as it cannot find a match in the history.

3.2 Chunk feature neural networks

We define the high-dimensional point cloud as *absolute* scene description. We notice building structures (walls, floors...) and large objects (desk, sofa...) usually do not update over a short period of time and thus do not need to be considered repeatedly. Therefore, we define what has changed in the scene as *relative* scene description, which can then be used as features to train the machine learning model.

In our case, we use a two-dimensional vector (d, l) as the *relative* scene description, where d represents the geometry changes and is calculated by the Chamfer Distance [51] of point clouds between its current and its first capture in history; l represents the illumination and equals the average illuminance of a given chunk.

For simplicity, we used a single-layer perceptron with ReLU activations. The model is trained with the Adam Optimizer on a cosine annealing learning rate starts on 0.0001 and a batch size of 4 with 4096 epochs.

3.3 Simulator

As a preliminary step, we evaluate the system in a modified VirtualHome [34] simulator, which provides the data needed for training the machine learning model and substitutes for the real world in the user study. We modified VirtualHome’s existing algorithm to produce IoT states, point clouds, and RGB files for all 20 cameras

in the room. To generate coherent daily activities, we designed a generator that randomly selects and combines tasks from the data that Puig et al. [34] collected. Since not all tasks are supported by their simulator, we also provided scripts to filter out incompatible ones. We introduced probability into the random process so that each IoT device has the same chance of being triggered. 11 IoTs are included in the simulator: five lights, three lamps, and three doors.

3.4 Programming interface

We developed an interface for HomeView to aid the IoT trigger-action programming [13] by presenting the context, which is considered essential [17]. We also built a baseline version with text descriptions for the user study. Both UIs were built using the GUI feature of the Open3D [51] in Python (Figure 4).

3.4.1 Context (History). This section allows the users to browse all timestamps where the actions under “When” get triggered. Once the user selects the specific timestamp, the corresponding IoT states will be visualized (HomeView) or shown as texts (baseline), and users can check the scenes before or after the trigger.

3.4.2 IoTs. This section is shown as either a 3D model (HomeView) or buttons and texts (baseline). Users can select an IoT by clicking on the corresponding button (baseline), or on the visualization in the 3D model (HomeView).

3.4.3 Automation. Inspired by the commercial softwares [3–5], our interface allows the users to program with If, Else, And, Or conditions.

3.4.4 Controls. This section is only available for HomeView, which allows the users to control properties related to the visualization. For example, *Remove ceiling* allow users to see the home interior.

4 EVALUATION

We present two early evaluations of HomeView: 1) visual fidelity comparison with the retrieve-from-history method 2) user feedback on programming IoTs using our system.

4.1 Visual fidelity

We compare the outcomes of HomeView with the ones from the retrieve-from-history method. We define a chunk to be *correct* if it is visually indistinguishable from the counterpart in the ground truth, which we quantified as $|\Delta d| \leq 0.01$ and $|\Delta l| \leq 0.05$. We define *PCC* as the percentage of correct chunks. retrieve-from-history method has $PCC = 12.25\%$, while HomeView’s PCC reaches 94.68%. Examples are given (Figure 3).

4.2 User feedback

We conducted user studies to evaluate how HomeView supports real-world IoT programming tasks by comparing it with the baseline (Figure 4).

4.2.1 Participants. Five participants (F=3, M=2), aged 21 to 24, were recruited. Two of them rarely or never used IoTs, one used IoTs daily, one used IoTs weekly, and one used IoTs a few times a year. Each session lasted for 50 minutes.



Figure 3: HomeView has better outcomes in terms of visual fidelity. Red boxes indicate incorrect chunks.

4.2.2 Tasks and Procedure. We designed three IoT programming tasks based on common routines people might have at home. The first task was to help the participants get familiar with the systems. The participants were asked to use either baseline or HomeView to complete the other two. The order in which participants were exposed to each system was counterbalanced using a Latin square (the 5th participant repeated the same order as the first one). We collected qualitative feedback throughout the session.

4.2.3 Results. Four out of five participants preferred HomeView over the baseline, saying it was much more intuitive and easier to understand. They said the major disadvantage of the baseline was the IoT names (“L1”, “D1” ...) did not make sense to them, and they could not map them into the real world in their heads. We asked if renaming the IoTs into “Bedroom Light” or “Kitchen Light” would solve the problem. One said:

That is definitely helpful, but I might have two identical IoTs in the same room, and I probably will still forget which one is which.

One participant who preferred the baseline system said that the UI is simpler, and clicking on buttons is much easier than clicking on 3D visualizations. This is due to the limitation of our current implementation. More specifically, three participants mentioned our 3D visualization is hard to rotate and click, indicating that a 2D floor plan is enough. Two participants with advanced coding

backgrounds said, since our simulator provides detailed information on the IoT states, they ended up programming solely based on that without even thinking about the meaning. One with an industrial design background mentioned a professional UI designer should be involved in improving our tool.

5 DISCUSSION

This section discusses the applications, limitations, and future work of HomeView.

5.1 Applications

HomeView helps users better visualize their homes, enabling them to control and program their IoTs remotely. The system is useful when the tenants are not the same as the person who set up their IoTs (operator), especially with children [8] or elderly users [45]. In those scenarios, HomeView’s ability to simulate historical situations allows the operator to have the context and feel situated when they program the automation.

HomeView is also helpful in commercial settings. Future office building tenants are likely to wear AR headsets while working in the environment. HomeView’s dynamic model is a great addition to the existing building information model [37], which contains building structures like walls and floors while missing furniture and objects placed by the tenants. Building managers can then better understand the tenants’ needs and program the building to reduce energy costs while not compromising the tenants’ experience.

5.2 Limitations and future work

Our system supports IoTs that cause visual differences in the scene, but we only tested lights, lamps, and doors due to time constraints. Meanwhile, we need to support IoTs that do not change the scene, for example, temperature sensors and humidity sensors. Future evaluations on those IoTs should be conducted.

Besides, we have limited experience with the HomeView model on real-world data, which is crucial as it is hard to address the limitations of an actual AR headset in the simulator. For example, in the real world, an AR headset only captures where the user faces instead of the entire environment. Therefore, we will field test the HomeView and run a more comprehensive user study that further demonstrate the capabilities of HomeView. We will then have an in-depth discussion on concerns such as privacy implications.

6 CONCLUSION

HomeView shows the promise of using AR headsets to automatically generate models that help users control, view, and configure IoT devices in the environments. Leveraging the sensing and communicating capabilities of AR headsets, future interaction with smart devices can become much easier and more intuitive.

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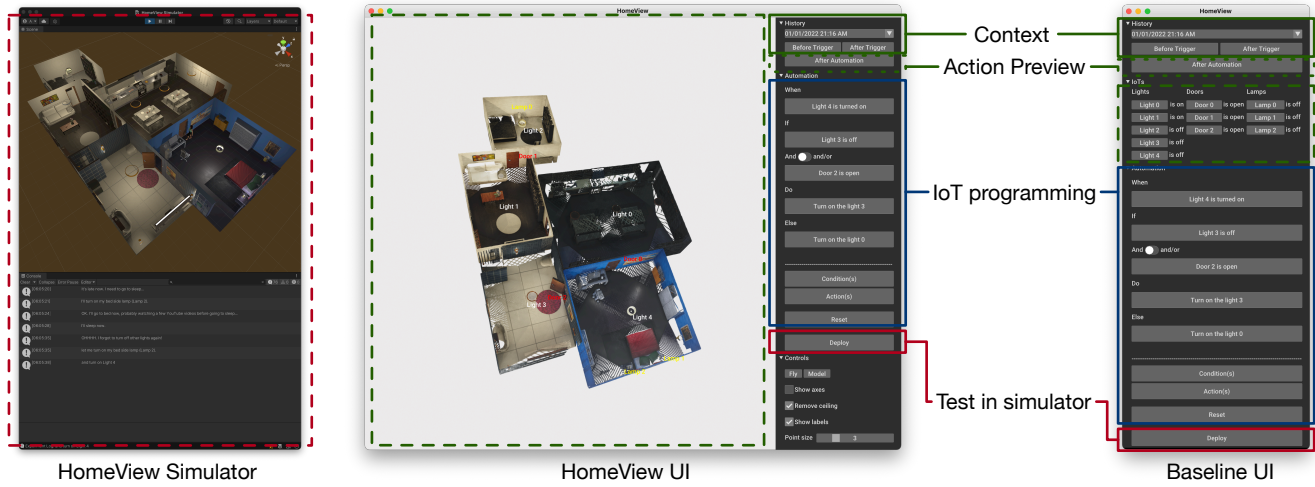


Figure 4: HomeView interface shows contextual information about the automation that helps the users program the IoTs.

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